

# **Prime Palindrome program**

**Title:** Prime Palindrome Number using Java

## **Objective / Aim :**

To write a Java program to check whether a given number is a Prime Palindrome or not.

## **System Requirements :**

### **Hardware:**

- Standard PC/Laptop
- Intel Core i3/i5 Processor

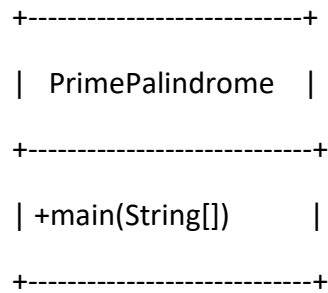
### **Software:**

- Windows/Linux/macOS
- JDK 17 or JDK 21
- Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt or Terminal

## **Algorithm / Logic Description :**

1. Start the program.
2. Read a number from the user.
3. Check whether the number is Prime.
4. Reverse the number.
5. Compare the reversed number with the original number.
6. If the number is both Prime and Palindrome, display "Prime Palindrome".
7. Otherwise, display "Not a Prime Palindrome".
8. Stop the program.

## UML Class Diagram :



## Source Code :

```
Package core_java;

import java.util.Scanner;

public class PrimePalindrome {

    public static void main(String[] args) {

        //Create Scanner object for input

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int count = 0;

        // Check all numbers from 1 to num to find the factors

        for (int i = 1; i <= num; i++) {

            if (num % i == 0)

                count++;

        }

        int temp = num;

        int rev = 0;

        // Reverse the number

        while (temp > 0) {
```

```

        int rem = temp % 10;

        rev = rev * 10 + rem;

        temp = temp / 10;
    }

    if (count == 2 && num == rev)

        System.out.println("Prime Palindrome");

    else

        System.out.println("Not a Prime Palindrome");

    sc.close();

}
}

```

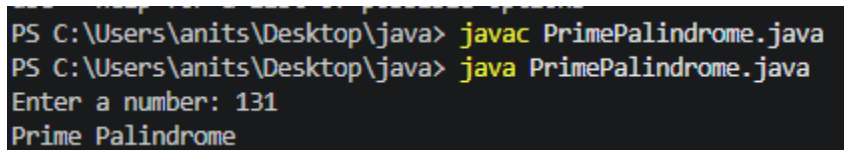
### Compilation & Execution Commands :

**Save the file as:** PrimePalindrome.java

**Compile:** javac PrimePalindrome.java

**Run:** java PrimePalindrome

### Output Screenshot :



```

PS C:\Users\anits\Desktop\java> javac PrimePalindrome.java
PS C:\Users\anits\Desktop\java> java PrimePalindrome.java
Enter a number: 131
Prime Palindrome

```

### Result :

Thus, the Java program to check whether a given number is a Prime Palindrome or not was successfully implemented and executed.